Standard Operating Procedure (SOP) for Telecom EDI 811 Implementation and Processing Using Altova MapForce & FlowForce Server

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Using Altova MapForce & FlowForce Server

Prepared For: Telecom EDI 811 Implementation R&D and Development Planning

Prepared Based On:

- Existing EDI 810 Implementation
- Existing MapForce (.mfd) Mappings
- ApexEDI Database Structure
- AT&T EDI 811 Implementation Guide Version 004010
- Actual Telecom Vendor 811 Transaction Set Analysis

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1. PURPOSE OF THIS DOCUMENT

This document explains:

1. What is EDI 811
2. Difference between EDI 810 and EDI 811
3. What can be reused from the current 810 implementation
4. What new development is required for 811
5. Important telecom segments
6. Database impact
7. MapForce development approach
8. FlowForce automation process
9. 997 acknowledgment requirement
10. Risks, validations, and implementation recommendations

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2. CURRENT ENVIRONMENT – EDI 810

Currently the system is processing EDI 810 Utility Invoice files using:

- Altova MapForce
- Altova FlowForce Server
- SQL Server (ApexEDI)
- FTP/SFTP Integration
- 997 Functional Acknowledgment

Current 810 Process Flow

```
FTP Pickup
   ↓
EDI 810 Validation
   ↓
MapForce Mapping
   ↓
Database Insert
   ↓
997 Generation
   ↓
Archive & Notification
```

Current Database Tables Used

- InvoiceHeader
- Invoice_Detail
- Invoice_Summary
- REF
- SLN
- PID
- PER
- MEA

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3. WHAT IS EDI 811?

EDI 811 is called:

> Consolidated Service Invoice / Statement

It is mainly used for:

- Telecom billing
- Service billing
- Usage processing
- Monthly recurring charges
- Itemized telecom billing
- Call detail processing
- Service-level charges
- Discounts and adjustments

According to the AT&T implementation guide, the EDI 811 transaction is used for:

"billing or reporting of complex and structured service invoice detail."

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4. MAJOR DIFFERENCE BETWEEN 810 AND 811

EDI 810	EDI 811
Utility invoice	Telecom/service billing
Semi-flat structure	Hierarchical structure
Simple invoice detail	Usage-level billing
Small-to-medium files	Very large files
Limited looping	Heavy nested looping
SAC charge handling	SLN/usage/telecom charge handling
Simple REF usage	Telecom identifiers and references
Simple validations	Advanced telecom validations

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5. MOST IMPORTANT DIFFERENCE

The biggest technical difference is:

EDI 811 uses Hierarchical Structure (HL Loops)

Unlike 810, Telecom 811 heavily uses:

- HL loops
- Nested hierarchy
- SLN loops
- Usage processing
- Telecom service identifiers
- Grouped billing structure

Example:

```
HL~1~~1~1
HL~2~1~4~1
HL~3~2~6~1
HL~4~3~8~0
```

Hierarchy Meaning:

```
Provider
   ↓
Billing Group
   ↓
Service
   ↓
Charge Detail
```

This is the MOST IMPORTANT development area in 811.

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6. WHAT CAN BE REUSED FROM CURRENT 810 IMPLEMENTATION?

Reusable Components

Existing Component	Reusable in 811
FlowForce Jobs	YES
FTP/SFTP Process	YES
Notification Framework	YES
Envelope Logic	YES
Database Connectivity	YES
Error Handling	YES
Archive Process	YES
997 Generation Logic	YES
REF Processing	PARTIAL
Validation Logic	PARTIAL

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7. WHAT NEW DEVELOPMENT IS REQUIRED FOR 811?

Major New Development Areas

Area	Complexity
HL Hierarchy Handling	HIGH
Nested Loop Processing	HIGH
SLN Processing	HIGH
Telecom Usage Processing	HIGH
Large File Optimization	HIGH
Usage Aggregation	HIGH
Telecom Identifiers	MEDIUM
BAL/QTY/AMT Handling	MEDIUM
Telecom Tax Logic	MEDIUM

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8. IMPORTANT SEGMENTS REQUIRED IN EDI 811

Segment	Purpose	Importance
HL	Hierarchical structure	MOST CRITICAL
LX	Line numbering	Important
SI	Telecom identifiers	VERY IMPORTANT
SLN	Telecom subline detail	MOST CRITICAL
BAL	Balance summary	Important
QTY	Usage quantity	Important
ITA	Taxes and adjustments	Important
PER	Contact information	Important
NM1	Organization/subscriber	Important
AMT	Monetary amount	Important

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9. HL SEGMENT EXAMPLE

Purpose

Defines telecom hierarchical structure.

Example

```
HL~1~~1~1
```

Meaning

- First hierarchy level
- Used for telecom billing hierarchy
- Controls parent-child loop processing

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10. LX SEGMENT EXAMPLE

Purpose

Used for telecom line sequence numbering.

Example

```
LX~1
```

Meaning

- First telecom billing line
- Used for service grouping

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11. SI SEGMENT EXAMPLE

Purpose

Stores telecom service identifiers.

Example

```
SI~TI~EN~1717924093078
```

Meaning

Value	Meaning
TI	Telecom Identifier
EN	Equipment/Service Number
1717924093078	Service Identifier

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12. SLN SEGMENT EXAMPLE

Purpose

Stores telecom subline billing detail.

Example

```
SLN~57~~I~1~M4~5~~I
```

Meaning

- Telecom charge detail
- Usage-level billing
- Subline charge processing

---

13. BAL SEGMENT EXAMPLE

Purpose

Stores invoice balance details.

Example

```
BAL~M~TT~2671.88
```

Meaning

Value	Meaning
TT	Total Due
2671.88	Invoice Balance

---

14. QTY SEGMENT EXAMPLE

Purpose

Stores telecom usage quantities.

Example

```
QTY~94~1
```

Meaning

- Usage quantity
- Service usage calculation

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15. ITA SEGMENT EXAMPLE

Purpose

Stores taxes, discounts, and telecom surcharges.

Example

```
ITA~N~TI~~ZZ~1057~~-5000~~100
```

Meaning

- Telecom discount/tax information
- Adjustment processing

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16. PER SEGMENT EXAMPLE

Purpose

Stores billing contact information.

Example

```
PER~BI~~TE~8003581111
```

Meaning

Value	Meaning
BI	Billing Contact
TE	Telephone
8003581111	Contact Number

---

17. NM1 SEGMENT EXAMPLE

Purpose

Stores telecom organization information.

Example

```
NM1~SJ~2~AT&T-ABN
```

Meaning

- Telecom provider information
- Organization identification

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18. AMT SEGMENT EXAMPLE

Purpose

Stores monetary amount information.

Example

```
AMT~SC~2671.88
```

Meaning

Value	Meaning
SC	Service Charge
2671.88	Amount

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19. IMPORTANT TELECOM QUALIFIERS

Qualifier	Meaning
12	Billing Account
AP	Accounts Receivable
TN	Telephone Number
CT	Contract Number
SV	Service Type

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20. CURRENT DATABASE IMPACT

Existing Tables Reusable

Table	Reusable
InvoiceHeader	YES
Invoice_Summary	YES
REF	YES
PID	YES
PER	YES

Existing Tables Requiring Enhancement

Table	Reason
Invoice_Detail	Telecom hierarchy
SLN	Deep telecom detail
REF	Telecom references

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21. NEW TABLES RECOMMENDED

Table	Purpose
ServiceDetail	Telecom services
UsageDetail	Usage records
TelecomCharges	Charge detail
CircuitReference	Telecom identifiers
CallDetail	Itemized call records

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22. MAPFORCE DEVELOPMENT APPROACH

Step 1 – Import 811 Schema

Import:

- X12 811 Version 004010

Step 2 – Import Database Structure

Import:

- SQL target tables
- Existing ApexEDI structure

Step 3 – Build Envelope Logic

Map:

- ISA
- GS
- ST
- SE
- GE
- IEA

Step 4 – Build Header Logic

Map:

- BIG
- REF
- N1
- DTM

Step 5 – Build HL Hierarchy Logic

MOST IMPORTANT DEVELOPMENT AREA.

Need:

- Parent-child hierarchy
- Nested looping
- Context handling
- Group processing

Step 6 – Build IT1 & SLN Logic

Need:

- Telecom billing processing
- Charge-level hierarchy
- Usage processing
- SLN charge detail

Step 7 – Build Validation Logic

Need:

- Mandatory segment validation
- TDS balancing
- Duplicate prevention
- Usage validation
- Telecom hierarchy validation

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23. IMPORTANT MAPFORCE FUNCTIONS REQUIRED IN 811

Function / Logic	Current 810	Needed in 811
if-else	YES	YES (more complex) **(Conditional logic)**
concat	YES	YES (Formatting)
exists	YES	YES (Validation)
equal	YES	YES
filter	YES	YES (heavy usage) **(Telecom filtering)**
remove-context	Moderate	VERY IMPORTANT **(Nested loop handling)**
sum	YES	YES **(Charge aggregation)**
trim	YES	YES
distinct-values	Limited	VERY IMPORTANT **(Duplicate prevention)**
group-by	Minimal	VERY IMPORTANT **(Usage grouping)**
nested loop handling	Medium	VERY HIGH
aggregation logic	Moderate	HIGH
usage calculations	Minimal	VERY HIGH
duplicate handling	Moderate	HIGH
hierarchy processing	Low	VERY HIGH
performance optimization	Moderate	CRITICAL

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24. FLOWFORCE AUTOMATION PROCESS

```
FTP/SFTP Pickup
        ↓
Local Incoming Folder
        ↓
EDI 811 Validation
        ↓
MapForce Execution
        ↓
Database Insert
        ↓
```

```
Generate 997
        ↓
Move ACK to Outgoing
        ↓
Archive Files
        ↓
Error Notification
```

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25. 997 ACKNOWLEDGMENT REQUIREMENT

According to the AT&T 811 implementation guide:

- 997 acknowledgment is REQUIRED
- AT&T expects 997 within 48 hours
- Connectivity testing includes 997 processing

Standard 997 Example

```
ST*997*0001~
AK1*CI*6049~
AK2*811*000013004~
AK5*A~
AK9*A*1*1*1~
SE*7*0001~
```

Meaning

- File accepted successfully

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26. PERFORMANCE CONSIDERATIONS

Telecom 811 files can become very large because of:

- Usage detail
- Itemized calls
- Multiple hierarchy levels
- Nested loops
- Large billing records

Need:

- Loop optimization
- Efficient grouping
- SQL indexing
- Archive cleanup
- Batch optimization

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27. RISKS & CHALLENGES

Risk	Mitigation
Large file size	Performance optimization
Nested hierarchy	Proper HL logic
Duplicate records	distinct-values()
Wrong aggregation	group-by validation
Usage mismatch	QTY validation
Telecom complexity	Extensive testing

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28. TESTING APPROACH

Unit Testing

- Segment-level testing

SIT Testing

- End-to-end telecom testing

UAT Testing

- Vendor validation

Performance Testing

- Large-file testing
- Multiple transaction testing

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29. FINAL TECHNICAL CONCLUSION

The current EDI 810 implementation provides a strong reusable framework for:

- Envelope processing
- FlowForce automation
- Database integration
- Notifications
- Error handling
- FTP/SFTP automation
- 997 acknowledgment processing

However, Telecom EDI 811 implementation requires moderate-to-high enhancement in:

- HL hierarchy handling
- Nested loop processing
- Usage-level billing
- SLN processing
- Telecom identifier handling
- Usage aggregation
- Large-file optimization
- Telecom validation logic

The actual vendor 811 transaction file confirms that telecom billing is significantly more hierarchical and usage-oriented compared to current 810 utility invoice processing.

Final Recommendation

A Proof-of-Concept (POC) implementation is strongly recommended before full production deployment.

Recommended implementation phases:

1. Connectivity & 997 Testing
2. HL Hierarchy Development
3. Telecom Segment Mapping
4. Database Enhancement
5. Performance Optimization
6. End-to-End Vendor Testing
7. Production Deployment